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Inventor(s)	Sugino; Takumi

Tire valve

Abstract

A tire valve includes: a valve stem passed through a valve mount hole of a rim and screwed with a nut, a retaining part extending sideways from a proximal end of the valve stem, an electronic device having a connection part that fits onto an outer side of the retaining part, a support ring set on the valve stem, the support ring and a portion of the connection part being sandwiched between the retaining part and an opening edge of the valve mount hole by fastening the nut, and an interference avoidance part formed by cutting off part of an outer circumferential surface of the support ring. The support ring and electronic device include guide parts which slide on each other by fastening the nut thereby causing the support ring and electronic device to approach to guide the support ring and electronic device toward the normal rotational position.

Inventors:	Sugino; Takumi (Ogaki, JP)
Applicant:	PACIFIC INDUSTRIAL CO., LTD. (Ogaki, JP)
Family ID:	83818885
Assignee:	PACIFIC INDUSTRIAL CO., LTD. (Ogaki, JP)
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Primary Examiner: Williams; Patrick C

Attorney, Agent or Firm: Oliff PLC

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a tire valve inserted into a valve mount hole extending through the rim of a tire wheel from an inner side of the rim, and screwed with a nut on an outer side.

BACKGROUND ART

(2) Tire valves of this type, with an electronic device attached to a proximal end of a valve stem, are commonly known (see, for example, Patent Document 1).

RELATED ART DOCUMENTS

Patent Documents

(3) Patent Document 1: JP 2004-276799 A (FIG. 2 and paragraph)

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(4) The above tire valve is desired to be mounted such that the electronic device is disposed in a predetermined normal rotational position around the valve mount hole relative to the tire wheel in consideration of centrifugal force, and the operation of attaching the tire valve to a tire wheel in this way required deft skills. Accordingly, the present disclosure proposes a technique that can facilitate the operation of attaching a tire valve such that an electronic device is set in a normal rotational position relative to a tire wheel.

Means of Solving the Problem

(5) A tire valve according to a first aspect of the present disclosure made to solve the above problem is a tire valve including: a valve stem passed through a valve mount hole extending through a rim of a tire wheel from an inner side of the rim and screwed with a nut on an outer side of the rim; a retaining part extending sideways from a proximal end of the valve stem; an electronic device having a connection part that fits onto an outer side of the retaining part; a support ring set on the valve stem, the support ring and a portion of the connection part being sandwiched between the retaining part and an opening edge of the valve mount hole when the nut is fastened; an interference avoidance part formed by cutting off part of an outer circumferential surface of the support ring and positioned adjacent to a wheel wall portion of the tire wheel near the valve mount hole; and guide parts provided to the support ring and the electronic device, wherein the guide parts slide on each other when the nut is fastened thereby causing the support ring and the electronic device to approach, to guide the support ring and the electronic device toward a normal rotational

position where the support ring and the electronic device are at correct rotational positions relative to each other around the valve stem, when the support ring and the electronic device are displaced from the normal rotational position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view of a tire wheel with a tire valve according to a first embodiment of the present disclosure attached thereto.
- (2) FIG. 2 is a cross-sectional side view of the tire valve seen from a second direction.
- (3) FIG. 3 is an exploded perspective view of the tire valve.
- (4) FIG. 4 is a cross-sectional side view of the tire valve in cross section A-A of FIG. 2.
- (5) FIG. 5(A) is a perspective view of an electronic device seen from above and (B) is a perspective view of the electronic device seen from below.
- (6) FIG. 6 is a perspective view with a cross section of the tire valve attached to a rim.
- (7) FIG. 7(A) is a perspective view of a support ring seen from a distal side, (B) is a perspective view of the support ring seen from a proximal side, and (C) is a cross-sectional side view of the support ring.
- (8) FIG. 8(A) is a perspective view and (B) is a cross-sectional side view of a support ring of a second embodiment.
- (9) FIG. 9 is a perspective view of an electronic device of a tire valve according to a third embodiment seen from below.
- (10) FIG. 10 is a cross-sectional side view of the tire valve.
- (11) FIG. 11(A) is a perspective view of a support ring seen from a distal side, (B) is a perspective view of the support ring seen from a proximal side, and (C) is a cross-sectional side view of the support ring.
- (12) FIG. 12(A) is a perspective view of a support ring according to a fourth embodiment seen from a distal side, and (B) is a perspective view of the support ring seen from a proximal side.
- (13) FIG. 13 is a cross-sectional side view of a tire valve according to a fifth embodiment.
- (14) FIG. 14 is a cross-sectional side view of a tire valve according to a sixth embodiment.
- (15) FIG. 15 is a perspective view of a support ring.
- (16) FIG. 16 is a perspective view of a support ring according to a seventh embodiment.
- (17) FIG. 17 is a cross-sectional side view of a tire valve according to an eighth embodiment.
- (18) FIG. 18(A) is a perspective view of a support ring seen from a distal side, (B) is a perspective view of the support ring seen from a proximal side, and (C) is a cross-sectional side view of the support ring.
- (19) FIG. 19 is a perspective view of a support ring according to a ninth embodiment.

MODE FOR CARRYING OUT THE INVENTION

First Embodiment

- (20) Hereinafter a tire valve **10A** according to a first embodiment of the present disclosure will be described with reference to FIG. 1 to FIG. 7. As shown in FIG. 1, the tire valve **10A** of this embodiment is configured to include an electronic device **50** at one end of a valve stem **11**, and attached to a tire wheel **90** such that the electronic device **50** is positioned inside the tire **93**, with the valve stem **11** passed through a valve mount hole **92** of the tire wheel.
- (21) The tire wheel **90** includes a rim forming part **95**, for example, on an outer side of a disc part **94**. The rim forming part **95** is configured to have a tubular part **91B** fitted and welded to an outer side of the disc part **94**, for example, with a pair of rims **91** extending sideways from both ends of the tubular part **91B** (FIG. 1 shows only one of the rims **91**). The rim **91** has a tire mount part **91A** at the distal end of a flange part **91C** that extends from an end portion of the tubular part **91B** and

bends sideways. The valve mount hole **92** extends through one of the rims **91** in the flange part **91C** near the tubular part **91B**. The flange part **91C** is slightly tilted outward. As shown in FIG. 2, a circumferential portion of the opening edge of the valve mount hole **92** is positioned in an inner corner between the flange part **91C** and the tubular part **91B**, and a notch **96** having a flat surface parallel to the axial direction of the valve mount hole **92**, for example, is formed in this inner corner.

(22) In this embodiment, the notch **96** in the tire wheel **90** described above corresponds to a “wheel wall portion” in the claims that is adjacent to an interference avoidance part **33** of a support ring **30A** to be described later. The “wheel wall portion” adjacent to the interference avoidance part **33** may not necessarily be formed by cutting off part of the tire wheel **90** and may be, for example, a portion of the tubular part **91B** that is not cut off. The tire wheel **90** may be made of iron or aluminum.

(23) The valve stem **11** is made of metal and, as shown in FIG. 3, configured in a symmetrical shape as a whole, with a cylindrical stem body **12** and a pair of retaining parts **13** extending from one end of the stem body to both sides. The valve stem **11** may be made of resin, and/or asymmetrical. Hereinafter in the description of various parts of the tire valve **10A**, the axial direction along the center axis of the stem body **12** shall be referred to as a first direction **H1**, and the direction in which the pair of retaining parts **13** extends shall be referred to as a second direction **H2**. One side in the first direction **H1** where there are the retaining parts **13** shall be referred to as a “proximal side”, and the opposite side as a “distal side”.

(24) As shown in FIG. 2, the stem body **12** accommodates a valve core **20** inside. The valve core **20** has a structure known as a check valve, which restricts the air from passing from the proximal side to the distal side in a flow passage **12R** inside the valve stem **11**, while allowing the air to pass in the opposite direction.

(25) As shown in FIG. 3, the stem body **12** on the outside includes a large-diameter part **12A** on the proximal side and a small-diameter part **12B** on the distal side, with a tapered part **12C** in the middle in the axial direction. The small-diameter part **12B** has an annular groove **12M1** in the middle in the axial direction, the distal side from this annular groove **12M1** being a threaded part **12G**. A valve cap **25** (see FIG. 1) that covers a distal end opening **12K** of the stem body **12** is screwed to this threaded part **12G**.

(26) The large-diameter part **12A** also has an annular groove **12M2** in the middle in the axial direction, the distal side from this annular groove **12M2** being a threaded part **12E**. As shown in FIG. 2, a nut **40** for fitting the valve stem **11** to the tire wheel **90** is screwed to the threaded part **12E**. The proximal side from the annular groove **12M2** of the large-diameter part **12A** is a fitting part **12F**, to which the support ring **30A**, a grommet **46**, and a washer **45** are fitted. When the tire valve **10A** is attached to the rim **91**, a middle portion of the fitting part **12F** extends through the valve mount hole **92** of the rim **91**, with the washer **45** abutted on the opening edge on the outer side of the rim **91** and the support ring **30A** abutted on the opening edge on the inner side of the rim **91**, and with the grommet **46** pressed into the valve mount hole **92** from the inner side of the rim **91** and making tight contact with an outer surface of the fitting part **12F** and an inner surface of the valve mount hole **92**. The shapes of the support ring **30A** and grommet **46** will be described in more detail later.

(27) As shown in FIG. 3, the pair of retaining parts **13** extends from the proximal end of the stem body **12** to both sides as described above. The pair of retaining parts **13** on the outside has an arcuate surface **13B** that is semicircular on the side facing the distal end of the stem body **12**. The pair of retaining parts **13** on the outside has a flat surface **13A** continuous over the pair of retaining parts **13** on the opposite side from the arcuate surface **13B**. The flow passage **12R** inside the stem body **12** opens at the center in the flat surface **13A**. Alternatively, the retaining parts **13** may be columnar so that it has a cylindrical outer surface as a whole.

(28) The electronic device **50** includes, for example, a wireless circuit and a sensor that detects a

condition of the tire **93**, and wirelessly transmits the detection results of the sensor to a tire monitoring device in the vehicle body (not shown). The tire monitoring device monitors the presence or absence of an abnormality in the condition of the tire **93** based on the received detection results. Concrete examples of the sensor include a pressure sensor that detects internal pressure of the tire **93**, a temperature sensor that detects temperature inside the tire **93**, an acceleration sensor capable of detecting vibration applied to the tire wheel **90**, and so on. Other sensors may be used. There may be one sensor, or a plurality of sensors.

(29) The electronic device **50** is configured such that an electrical circuit including the sensor and wireless circuit described above is packaged in a resin-made housing **50H**. The housing **50H** includes a case part **51** that accommodates the electrical circuit, and a connection part **52A** for coupling the electronic device **50** to the valve stem **11**. The case part **51** is elongated along the second direction **H2** and flat in a direction perpendicular to the second direction **H2**. Hereinafter the direction in which the case part **51** of the electronic device **50** is flattened shall be referred to as an up-down direction, one side being the upper side and the other side being the lower side. The direction perpendicular to both of the direction in which the case part **51** is flattened and the second direction **H2** shall be referred to as a front-back direction of the electronic device **50**, one side being the front side and the other side being the rear side.

(30) While the lower surface of the case part **51** is flat as shown in FIG. 5(B), the upper surface of the case part **51** is uneven as shown in FIG. 5(A). The connection part **52A** as a whole has a trough shape with an open top, and is connected to the case part **51** on the front and upper side thereof.

(31) To be more specific, the connection part **52A** includes a curved wall **53** extending forward from a lower edge portion on the front face of the case part **51** and arcuately curved upward, and a pair of connecting walls **55** connecting both ends in the second direction **H2** of the curved wall **53** with the front face and an upper front portion of the case part **51**. The pair of connecting walls **55** is located near one end and the other end in the second direction **H2** of the case part **51**. A pair of L-shaped legs **51L** extends out in the second direction **H2** from the surfaces of the pair of connecting walls **55** on the opposite sides to the surfaces facing each other. The pair of L-shaped legs **51L** extends from the case part **51** to points spaced away therefrom on both sides in the second direction **H2**, where they bend downwards, the distal ends being positioned lower than the lower surface of the case part **51**. The pair of L-shaped legs **51L** is also integral with the front face of the case part **51**.

(32) A slot **53N** is formed at the center in the second direction **H2** of the curved wall **53**, extending from near the lower end to near the upper end. The stem body **12** is passed through the slot **53N**, so that the pair of retaining parts **13** of the valve stem **11** fits inside a pair of arcuate grooves **54** that is part of the curved wall **53** with the slot **53N** disposed therebetween.

(33) As shown in FIG. 6, the entire inner face of the pair of arcuate grooves **54** except for upper and lower ends is an inner arcuate surface **54A** that is an arcuate surface having the same radius of curvature as that of the arcuate surface **13B** of the pair of retaining parts **13**. The entire outer face of the pair of arcuate grooves **54** except for upper and lower ends is an outer arcuate surface **54B** that is an arcuate surface having the same center as that of the inner arcuate surface **54A**. Moreover, as shown in FIG. 5(B), a pair of metal arcuate discs **56** is embedded in the pair of arcuate grooves **54**. Each arcuate disc **56** is located near the slot **53N** in each arcuate groove **54** in an arcuate shape extending from the upper end to the lower end of the arcuate groove **54**, and has an inner face forming part of the inner arcuate surface **54A** and an outer face forming part of the outer arcuate surface **54B**. As shown in FIG. 6, the inner face of the arcuate disc **56** makes contact with the arcuate surface **13B** of the retaining parts **13** as part of the inner arcuate surface **54A**. A plurality of through holes **56A** is formed all over the arcuate discs **56**.

(34) The pair of arcuate discs **56** is embedded in the pair of arcuate grooves **54** by insertion molding. Instead, for example, the housing **50H** may be formed such as to have a pair of slits in the pair of arcuate grooves **54**, and the pair of arcuate discs **56** may be inserted into the pair of slits.

Alternatively, the entire curved wall **53** may be made of metal, and this metal part forming the curved wall **53** may be embedded in the housing **50H** by insertion molding. Alternatively, the arcuate grooves **54** may not contain any metal parts and made only of resin (including reinforced resin reinforced with glass fiber, for example).

(35) As shown in FIG. 5(B), a pair of opposing surfaces of the pair of arcuate grooves **54** on both sides of the slot **53N** is formed with a pair of tapered guide surface **57**, which corresponds to a “guide part” in the claims. To be more specific, the opening edge where the hole inner face of the slot **53N** and the outer face of the curved wall **53** intersect is chamfered to a mid-point in the direction in which the slot extends through, for example. Part of this chamfered surface **53M** positioned along the pair of arcuate grooves **54** serves as the pair of tapered guide surfaces **57**. The pair of tapered guide surfaces **57** is sloped toward each other from the outer face to the inner face of the pair of arcuate grooves **54**. The hole inner face of the slot **53N** on the side closer to the inner face of the curved wall **53** from the mid-point in the direction in which the slot extends through is parallel to the radial direction of the circular arc of the inner arcuate surface **54A** and outer arcuate surface **54B**.

(36) FIG. 7 shows the support ring **30A** mentioned above as a single unit. The support ring **30A** is made of metal, for example, its center axis being parallel to the first direction **H1**. As shown in FIG. 6 and as mentioned above, the support ring is fitted onto the valve stem **11** and disposed on the inner side of the rim **91**. Namely, the support ring is sandwiched between the rim **91** and the connection part **52A** of the electronic device **50**.

(37) To be more specific, as shown in FIG. 7(B), the support ring **30A** has a flat abutment surface **36** perpendicular to the axial direction in an axial mid-point on the outside. An outer surface of the support ring **30A** on the distal end (rim **91**) side of the abutment surface **36** is a large-diameter part **31**, and an outer surface on the proximal end (electronic device **50**) side of the abutment surface **36** is a small-diameter part **34**. In an axial mid-point inside the large-diameter part **31** is a tapered surface **31M** as shown in FIG. 7(C). An inner surface of the support ring **30A** on the proximal end side of the tapered surface **31M** is a small-diameter hole **34A**, and an inner surface on the distal end side of the tapered surface **31M** is a large-diameter hole **31A**. Moreover, as shown in FIG. 7(A), the distal end of the support ring **30A** has chamfers **31C** and **31D** respectively on the outer side and inner side. As shown in FIG. 7(B), the proximal end of the support ring **30A** has a chamfer **34C** on the outer side.

(38) As shown in FIG. 7(A), a pair of interference avoidance parts **33** having a flat surface **33A** parallel to the first direction **H1** of the large-diameter part **31** is provided on the outer face of the large-diameter part **31** by cutting off two parts circumferentially spaced apart by 180°. The flat surface **33A** of the pair of interference avoidance parts **33** extends in the first direction **H1** over the entire large-diameter part **31**, and extends in the radial direction of the large-diameter part **31** as far as to the chamfer **31D** on the inner side of the large-diameter part **31**, as a result of which a pair of arcuate notches **33K** is formed in distal end parts of the large-diameter part **31** where the pair of interference avoidance parts **33** is formed.

(39) As shown in FIG. 7(B), a pair of guide slopes **35**, which is flat surfaces inclined to the first direction **H1** of the small-diameter part **34**, is formed on the outer face of the small-diameter part **34** by cutting off two parts circumferentially spaced apart by 180°. The pair of guide slopes **35** corresponds to “guide parts” in the claims, and extends from near the abutment surface **36** on the outside of the small-diameter part **34** to the distal end face of the small-diameter part **34**. As shown in FIG. 4, the pair of guide slopes **35** is inclined to the first direction **H1** at substantially the same angle as the angle of inclination of the pair of tapered guide surfaces **57** relative to the radial direction of the circular arc of the inner arcuate surface **54A** and outer arcuate surface **54B** of the electronic device **50** described above (up-down direction in FIG. 4). Moreover, as shown in FIG. 7(B), the pair of guide slopes **35** and the pair of interference avoidance parts **33** are disposed 90° out of alignment in the circumferential direction of the support ring **30A**, the support ring **30A** as a

whole having a 180° rotation symmetry.

(40) When the nut **40** is not fastened on the threaded part **12E** of the valve stem **11**, the support ring **30A** can rotate around the valve stem **11** relative to the electronic device **50**. The position where the direction in which the pair of guide slopes **35** of the support ring **30A** is aligned matches the direction in which the pair of arcuate grooves **54** of the electronic device **50** is aligned as shown in FIG. **4** is the normal rotational position where the support ring **30A** and electronic device **50** are correctly positioned relative to each other. When the support ring **30A** and electronic device **50** are in this normal rotational position and the support ring **30A** is pressed in the first direction **H1** toward the electronic device **50**, the abutment surface **36** of the support ring **30A** makes contact with the outer arcuate surfaces **54B** of the pair of arcuate grooves **54**, the small-diameter part **34** of the support ring **30A** is received between the pair of arcuate grooves **54**, and the pair of guide slopes **35** becomes adjacent to the pair of tapered guide surfaces **57** of the electronic device **50** with a slight gap therebetween. In this state, the pair of arcuate discs **56** embedded in the pair of arcuate grooves **54** is sandwiched between the pair of retaining parts **13** of the valve stem **11** and the support ring **30A**.

(41) The grommet **46**, which is made of elastomer, is shown in FIG. **2** in a state before being compressed, and in FIG. **4** in a state of being compressed. The grommet **46** before being compressed has an inner face and an outer face with a uniform diameter as shown in FIG. **2**, and a distal end face and a proximal end face that are perpendicular to the first direction **H1**, with an annular protrusion **46A** protruding from an inner edge on the distal end face. The annular protrusion **46A** is tapered on the outside. The grommet **46** is fitted into the large-diameter hole **31A** of the support ring **30A**, with the proximal end face abutted on the tapered surface **31M**, and the distal end face positioned more forward than the support ring **30A**.

(42) The structure of the tire valve **10A** according to this embodiment has been described above. This tire valve **10A** is attached to the tire wheel **90** as described below. The tire valve **10A**, with the nut **40** and washer **45** removed from the valve stem **11**, is inserted into the valve mount hole **92** of the tire wheel **90** with no tire **93** attached thereto, from the inner side of the rim **91**. The washer **45** and nut **40** are then mounted to the valve stem **11** protruding from the valve mount hole **92** to the outer side of the rim **91**, and the nut **40** is fastened lightly (i.e., temporarily).

(43) At this time, the annular protrusion **46A** of the grommet **46** is received into the valve mount hole **92** from the inner side of the rim **91**. One of the interference avoidance parts **33** of the support ring **30A** is set adjacent to the notch **96** of the tire wheel **90**. Further, the lower side of the electronic device **50** is set to face the tubular part **91B** of the tire wheel **90**. Since the electronic device **50** is tiltable around the retaining part **13** of the valve stem **11** at this stage, it is adjusted to take a prescribed normal tilted attitude. Specifically, in this embodiment, the lower side of the electronic device **50** is parallel to the rotation center of the tire wheel **90** in the normal tilted attitude, for example. The electronic device is adjusted to take such an attitude, after which the lower faces of the pair of L-shaped legs **51L** are abutted on an outer circumferential surface of the tubular part **91B** of the tire wheel **90**.

(44) When the nut **40** is fastened with a prescribed fastening torque (final fastening), the support ring **30A** and the pair of arcuate grooves **54** of the electronic device **50** are sandwiched between the pair of retaining parts **13** of the valve stem **11** and the inner face of the rim **91**, so that the electronic device **50** is fixed to a certain tilted attitude. The grommet **46** is compressed between the support ring **30A** and the rim **91** to make tight contact with an inner surface of the valve mount hole **92**, an opening edge of the rim **91**, an outer surface of the fitting part **12F** of the valve stem **11** as shown in FIG. **4**, so that the valve mount hole **92** is sealed. The above procedure completes the mounting of the tire valve **10A**.

(45) When the tire valve **10A** is passed through the valve mount hole **92**, before the nut **40** is fastened, the support ring **30A** is set to a certain rotational position relative to the valve mount hole **92** by the interference avoidance parts **33** and the notch **96** being adjacent to each other. In this

state, however, a situation can arise where the electronic device **50** is displaced from the normal rotational position relative to the support ring **30A**. When this is the case, according to the tire valve **10A** of this embodiment, the electronic device **50** is guided to the normal rotational position relative to the support ring **30A** in the process in which the nut **40** is fastened to the valve stem **11**, causing the support ring **30A** to approach the connection part **52A** of the electronic device **50**, and in which the pair of guide slopes **35** of the support ring **30A** and the pair of tapered guide surfaces **57** of the electronic device **50** slide on each other, i.e., the displacement from the normal rotational position is automatically corrected. The tire valve **10A** of this embodiment has such an automatic alignment function whereby the electronic device **50** is automatically guided to the normal rotational position in the process in which the tire valve **10A** is attached to the tire wheel **90**, so that the operation of attaching the tire valve **10A** is made easier.

(46) When the support ring **30A** and electronic device **50** are largely displaced from each other more than a maximum tolerable misalignment angle (of, for example 30° in this embodiment), below which the support ring **30A** and the electronic device **50** can be guided toward the normal rotational position, part of the support ring **30A** without the guide slopes **35** abuts on the pair of tapered guide surfaces **57** of the electronic device **50** and prohibits the support ring **30A** and electronic device **50** from approaching each other. This allows the operator to realize that the support ring **30A** and electronic device **50** are largely displaced from the normal rotational position, and prevents a situation where the support ring **30A** and electronic device **50** are left misaligned relative to each other.

(47) Since the support ring **30A** including the pair of interference avoidance parts **33** and the pair of guide slopes **35** has a rotation symmetry as a whole, it is easy to put the support ring **30A** and electronic device **50** back to the normal rotational position by turning them in suitable directions relative to each other, when the support ring **30A** and electronic device **50** are largely displaced from each other more than the maximum tolerable misalignment angle, below which the support ring **30A** and the electronic device **50** can be guided to the normal rotational position. The rotation symmetry of the support ring **30A** allows the electronic device **50** and support ring **30A** to be readily set in the normal rotational position when assembling the support ring **30A** by inserting the valve stem **11** into the ring, so that the operation of assembling the tire valve **10A** is also made easier.

(48) Since the electronic device **50** is tiltable around an axis perpendicular to the axial direction of the valve stem **11** (which is also the axial direction of the valve mount hole **92**), the tire valve **10A** can be fixed to various types of tire wheels **90** having a valve mount hole **92** in different positions such that the electronic device **50** does not interfere with an inner face of the tire wheel **90** by adjusting the tilted attitude of the electronic device as required.

(49) Since the pair of metal arcuate discs **56** is embedded in the pair of arcuate grooves **54** of the resin-made housing **50H** of the electronic device **50**, the pair of metal arcuate discs **56** is held between the pair of retaining parts **13** of the valve stem **11** made of metal and the support ring **30A** made of metal when the electronic device **50** is fixed in any desired tilted attitude, which provides a good balance in strength and allows the electronic device **50** to be stably fixed in the desired tilted position.

(50) When the electronic device **50** is located at the normal rotational position relative to the support ring **30A**, the pair of tapered guide surfaces **57** of the electronic device **50** and the pair of guide slopes **35** of the support ring **30A** provided for the automatic alignment function described above face each other with a gap therebetween, which prevents a large force from being applied to the pair of guide slopes **35** and the pair of tapered guide surfaces **57** when the nut **40** is fastened.

Second Embodiment

(51) This embodiment is shown in FIG. **8** and has a support ring **30B** with a different structure from that of the support ring **30A** of the tire valve **10A** of the first embodiment. Namely, the support ring **30A** of the first embodiment has the small-diameter part **34** on the proximal end side of the

abutment surface **36** as shown in FIG. **7**, with the pair of guide slopes **35** formed on this small-diameter part **34**. The support ring **30B** of this embodiment does not have a small-diameter part **34** as shown in FIG. **8** and instead, a lug **37** extends out from part of the opening edge of the abutment surface **36**, with one guide slope **35** formed on this lug **37**. The interference avoidance part **33** is provided only at one location 90° away from the guide slope **35**. Other configurations are the same as those of the first embodiment.

(52) This support ring **30B** used for the tire valve **10A** of the first embodiment can also provide the automatic alignment function described in the first embodiment and makes the operation of attaching the tire valve **10A** to the tire wheel **90** easier. The connection part **52A** of the electronic device **50** may be configured such that the tapered guide surface **57** is not formed to the arcuate groove **54** on the opposite side to the arcuate groove **54** having the tapered guide surface **57** that will abut the guide slope **35** of the support ring **30B** in the normal rotational position, so that the lug **37** will interfere with the arcuate groove **54** when the electronic device **50** is 180° out of alignment from the normal rotational position relative to the support ring **30B**.

Third Embodiment

(53) This embodiment is shown in FIG. **9** to FIG. **11** and different from the first embodiment in the configurations of a connection part **52C** of the electronic device **50** and a support ring **30C**.

Namely, as shown in FIG. **9**, the connection part **52C** of this embodiment has a pair of arcuate ribs **58** protruding from the outer arcuate surface **54B** of the pair of arcuate grooves **54**. The pair of arcuate ribs **58** extends along the circumferential direction of the outer arcuate surface **54B** from the lower end to the upper end of the arcuate grooves **54**. The pair of arcuate ribs **58** is located farther from the slot **53N** than the parts of the pair of arcuate grooves **54** where the pair of arcuate discs **56** is embedded. As shown in FIG. **10**, the opposing surfaces of the pair of arcuate ribs **58** form a pair of tapered guide surfaces **57C**. This connection part **52C** is not formed with the chamfered surface **53M** (see FIG. **5(B)**), part of which is the pair of tapered guide surfaces **57** described in the first embodiment along the opening edge of the slot **53N**.

(54) Corresponding to the connection part **52C** described above, the support ring **30C** of a tire valve **10C** in this embodiment does not have the small-diameter part **34** on the proximal side of the abutment surface **36**, the abutment surface **36** itself being the proximal end face, as shown in FIG. **11**. The support ring **30C** is formed with a pair of guide slopes **35C** at two locations each 90° away from each of the pair of interference avoidance parts **33** so that the support ring as a whole has a rotation symmetry. Moreover, the pair of guide slopes **35C** is positioned at the proximal end on the outer circumferential surface of the support ring **30C** and inclined to the axial direction of the support ring **30C** at the same angle as the angle of inclination of the pair of tapered guide surfaces **57C** relative to the radial direction of the outer arcuate surface **54B**. When the electronic device **50** is set in the normal rotational position relative to the support ring **30C**, the abutment surface **36** makes contact with a portion of the outer arcuate surface **54B** of the pair of arcuate grooves **54** between the pair of arcuate ribs **58**, and the pair of tapered guide surfaces **57C** becomes adjacent to the pair of guide slopes **35C**.

(55) The tire valve **10C** of this embodiment also provides the automatic alignment function described in the first embodiment and makes the operation of attaching the tire valve **10C** to the tire wheel **90** easier.

Fourth Embodiment

(56) This embodiment is shown in FIG. **12** and includes a support ring **30D** configured without one of the interference avoidance parts **33** and one of the guide slopes **35C** of the support ring **30C** in the tire valve **10C** of the third embodiment. Other configurations are the same as those of the third embodiment. The configuration of this embodiment also provides the automatic alignment function described in the first embodiment similarly to the second embodiment.

Fifth Embodiment

(57) A tire valve **10E** of this embodiment is shown in FIG. **13** and different from the third

embodiment in the configurations of a connection part **52E** of the electronic device **50** and a support ring **30E**. Namely, the connection part **52E** of this embodiment has the pair of arcuate ribs **58** located closer to the slot **53N** than the parts of the pair of arcuate grooves **54** where the pair of arcuate discs **56** is embedded, and has the pair of tapered guide surfaces **57E** on the opposite side to the opposing surfaces of the pair of arcuate ribs **58**.

(58) The support ring **30E** on the other hand has grooves **39** at the proximal end, extending in a direction in which the pair of interference avoidance parts **33** (not shown) is aligned (direction perpendicular to the paper surface of FIG. **13**). A pair of inner side faces of the grooves **39** forms a pair of guide slopes **35E** inclined to the axial direction of the support ring **30E**. When the electronic device **50** is set in the normal rotational position relative to the support ring **30E**, the abutment surface **36** makes contact with the outer arcuate surface **54B** of the pair of arcuate grooves **54**, and the pair of tapered guide surfaces **57E** becomes adjacent to the pair of guide slopes **35E**. Thus, the tire valve **10E** of this embodiment also provides the automatic alignment function described in the first embodiment.

Sixth Embodiment

(59) A tire valve **10F** of this embodiment is shown in FIG. **14** and FIG. **15**. As shown in FIG. **15**, a support ring **30F** of the tire valve **10F** of this embodiment has a pair of rotation restricting walls **35S** that is parallel to the axial direction and flat, instead of the pair of guide slopes **35** on the small-diameter part **34** of the support ring **30A** of the first embodiment. Correspondingly, a connection part **52F** is not formed with the chamfered surface **53M** (see FIG. **5(B)**), part of which is the pair of tapered guide surfaces **57**, along the opening edge of the slot **53N**.

(60) As shown in FIG. **14**, when the electronic device **50** is set in the normal rotational position relative to the support ring **30F**, the abutment surface **36** makes contact with the outer arcuate surface **54B** of the pair of arcuate grooves **54**, and a pair of rotation restricting walls **35S** becomes adjacent to opposing surfaces **54S** of the pair of arcuate grooves **54**. While the tire valve **10F** of this embodiment does not provide the automatic alignment function described in the first embodiment, the rotation restricting walls **35S** facing the opposing surfaces **54S** restrict the rotation of the electronic device **50** relative to the support ring **30F** and keeps the electronic device **50** stably in the normal rotational position.

Seventh Embodiment

(61) This embodiment is shown in FIG. **16**. Similarly to the relationship between the first embodiment and the second embodiment, the lug **37** protrudes from the abutment surface **36**, with the rotation restricting wall **35S** formed on this lug **37**, in exchange for elimination of the small-diameter part **34** from the support ring **30F** of the sixth embodiment, and a support ring **30G** is configured without one of the interference avoidance parts **33**. This configuration also provides similar effects as those of the sixth embodiment.

Eighth Embodiment

(62) A tire valve **10H** of this embodiment is shown in FIG. **17** and FIG. **18**. As shown in FIG. **17**, a connection part **52H** of the electronic device **50** in the tire valve **10H** of this embodiment is configured such that the opposing surfaces of the pair of arcuate ribs **58** in the connection part **52C** of the third embodiment are changed to a pair of rotation restricting walls **58S** that is parallel to each other and flat. A support ring **30H** on the other hand does not have the small-diameter part **34** described above as shown in FIG. **18**, the abutment surface **36** itself being the proximal end face, and instead is formed with interference avoidance parts **33** at four equally spaced locations on the outer circumferential surface. One of the interference avoidance parts **33** is set to face the notch **96** (see FIG. **2**) of the tire wheel **90**, and the flat surfaces **33A** of the pair of interference avoidance parts **33** on both sides of that interference avoidance part **33** are set adjacent to the rotation restricting walls **58S** of the pair of arcuate ribs **58** as shown in FIG. **17**. Similarly to the sixth embodiment, the configuration of this embodiment also restricts the rotation of the electronic device **50** relative to the support ring **30H** and provides the effect of keeping the electronic device

50 stably in the normal rotational position.

Ninth Embodiment

(63) This embodiment is shown in FIG. **19**. Similarly to the relationship between the third embodiment and the fourth embodiment, a support ring **30I** is configured such that two adjacent interference avoidance parts **33** are removed from the support ring **30H** of the eighth embodiment. This configuration also provides similar effects as those of the eighth embodiment.

(64) [Notes]

(65) While specific examples of the techniques included in the claims are disclosed in the specification and drawings, the techniques set forth in the claims are not limited to these specific examples but rather include various modifications and alterations of the specific examples, as well as partial extracts from the specific examples.

DESCRIPTION OF THE REFERENCE NUMERAL

(66) **10A, 10C, 10E, 10F, 10H** Tire valve **11** Valve stem **13** Retaining part **30A to 30I** Support ring **33** Interference avoidance part **34A** Small-diameter hole **35, 35C, 35E** Guide slope (guide part) **40** Nut **50** Electronic device **52A, 52C, 52E, 52F, 52H** Connection part **53** Curved wall **54** Arcuate groove **56** Arcuate disc **57, 57C, 57E** Tapered guide surface (guide part) **90** Tire wheel **91** Rim **92** Valve mount hole **96** Notch (wheel wall portion) **H1** First direction **H2** Second direction

Claims

1. A tire valve comprising: a valve stem passed through a valve mount hole extending through a rim of a tire wheel from an inner side of the rim, and screwed with a nut on an outer side of the rim; a retaining part extending sideways from a proximal end of the valve stem; an electronic device having a connection part that fits onto an outer side of the retaining part; a support ring set on the valve stem, the support ring and a portion of the connection part being sandwiched between the retaining part and an opening edge of the valve mount hole by fastening the nut; an interference avoidance part formed by cutting off part of an outer circumferential surface of the support ring and positioned adjacent to a wheel wall portion of the tire wheel near the valve mount hole; and guide parts provided to the support ring and the electronic device, wherein the guide parts slide on each other by fastening the nut thereby causing the support ring and the electronic device to approach, to guide the support ring and the electronic device toward a normal rotational position where the support ring and the electronic device are at correct rotational positions relative to each other around the valve stem, when the support ring and the electronic device are displaced from the normal rotational position.

2. The tire valve according to claim 1, wherein the guide part of one of the support ring and the electronic device interferes with a portion other than the guide part of an other one of the support ring and the electronic device to prohibit the support ring and the electronic device from approaching each other when the support ring and the electronic device are displaced from each other more than a maximum tolerable misalignment angle, below which the guide parts of the support ring and the electronic device are capable of guiding the support ring and the electronic device toward the normal rotational position by sliding on each other.

3. The tire valve according to claim 2, wherein the interference avoidance part is formed at two locations spaced apart by 180° on an outer circumferential surface of the support ring, the interference avoidance parts and the guide parts being formed in rotation symmetry to the support ring.

4. The tire valve according to claim 2, wherein the retaining part extends to both sides of the proximal end of the valve stem in a second direction perpendicular to a first direction that is an axial direction of the valve stem, and the connection part includes a pair of arcuate grooves that rotatably fits onto the retaining part on both sides of the valve stem, the pair of arcuate grooves being sandwiched in the first direction between the retaining part and the support ring by fastening

the nut so as to be stopped from rotating.

5. The tire valve according to claim 4, wherein the valve stem and the support ring are made of metal while the pair of arcuate grooves is made of resin, and a pair of metal arcuate discs is embedded in a portion of the pair of arcuate grooves sandwiched between the support ring and the retaining part, arcuate inner side faces and outer side faces of the pair of arcuate discs being exposed on arcuate inner side faces and outer side faces of the pair of arcuate grooves.

6. The tire valve according to claim 5, wherein the guide part of the electronic device includes a tapered surface formed on surfaces of the pair of arcuate grooves facing each other and sloped toward the valve stem in a direction toward the retaining part, and the guide part of the support ring includes a tubular protrusion received between the pair of arcuate grooves, and a pair of guide slopes formed at two 180° spaced apart locations on the tubular protrusion and sloped toward each other in a direction toward the retaining part, the guide slopes being parallel to a third direction perpendicular to the first direction and the second direction, the pair of guide slopes and the tapered surfaces being adjacent to or abutting each other when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

7. The tire valve according to claim 6 wherein a gap is formed between the pair of guide slopes and the pair of tapered surfaces when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

8. The tire valve according to claim 4, wherein the guide part of the electronic device includes a tapered surface formed on surfaces of the pair of arcuate grooves facing each other and sloped toward the valve stem in a direction toward the retaining part, and the guide part of the support ring includes a tubular protrusion received between the pair of arcuate grooves, and a pair of guide slopes formed at two 180° spaced apart locations on the tubular protrusion and sloped toward each other in a direction toward the retaining part, the guide slopes being parallel to a third direction perpendicular to the first direction and the second direction, the pair of guide slopes and the tapered surfaces being adjacent to or abutting each other when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

9. The tire valve according to claim 8, wherein a gap is formed between the pair of guide slopes and the pair of tapered surfaces when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

10. The tire valve according to claim 1, wherein the interference avoidance part is formed at two locations spaced apart by 180° on an outer circumferential surface of the support ring, the interference avoidance parts and the guide parts being formed in rotation symmetry to the support ring.

11. The tire valve according to claim 10, wherein the retaining part extends to both sides of the proximal end of the valve stem in a second direction perpendicular to a first direction that is an axial direction of the valve stem, and the connection part includes a pair of arcuate grooves that rotatably fits onto the retaining part on both sides of the valve stem, the pair of arcuate grooves being sandwiched in the first direction between the retaining part and the support ring by fastening the nut so as to be stopped from rotating.

12. The tire valve according to claim 11, wherein the valve stem and the support ring are made of metal while the pair of arcuate grooves is made of resin, and a pair of metal arcuate discs is embedded in a portion of the pair of arcuate grooves sandwiched between the support ring and the retaining part, arcuate inner side faces and outer side faces of the pair of arcuate discs being exposed on arcuate inner side faces and outer side faces of the pair of arcuate grooves.

13. The tire valve according to claim 11, wherein the guide part of the electronic device includes a tapered surface formed on surfaces of the pair of arcuate grooves facing each other and sloped toward the valve stem in a direction toward the retaining part, and the guide part of the support ring includes a tubular protrusion received between the pair of arcuate grooves, and a pair of guide slopes formed at two 180° spaced apart locations on the tubular protrusion and sloped toward each

other in a direction toward the retaining part, the guide slopes being parallel to a third direction perpendicular to the first direction and the second direction, the pair of guide slopes and the tapered surfaces being adjacent to or abutting each other when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

14. The tire valve according to claim 13, wherein a gap is formed between the pair of guide slopes and the pair of tapered surfaces when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

15. The tire valve according to claim 1, wherein the retaining part extends to both sides of the proximal end of the valve stem in a second direction perpendicular to a first direction that is an axial direction of the valve stem, and the connection part includes a pair of arcuate grooves that rotatably fits onto the retaining part on both sides of the valve stem, the pair of arcuate grooves being sandwiched in the first direction between the retaining part and the support ring by fastening the nut so as to be stopped from rotating.

16. The tire valve according to claim 15, wherein the valve stem and the support ring are made of metal while the pair of arcuate grooves is made of resin, and a pair of metal arcuate discs is embedded in a portion of the pair of arcuate grooves sandwiched between the support ring and the retaining part, arcuate inner side faces and outer side faces of the pair of arcuate discs being exposed on arcuate inner side faces and outer side faces of the pair of arcuate grooves.

17. The tire valve according to claim 16, wherein the guide part of the electronic device includes a tapered surface formed on surfaces of the pair of arcuate grooves facing each other and sloped toward the valve stem in a direction toward the retaining part, and the guide part of the support ring includes a tubular protrusion received between the pair of arcuate grooves, and a pair of guide slopes formed at two 180° spaced apart locations on the tubular protrusion and sloped toward each other in a direction toward the retaining part, the guide slopes being parallel to a third direction perpendicular to the first direction and the second direction, the pair of guide slopes and the tapered surfaces being adjacent to or abutting each other when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

18. The tire valve according to claim 17, wherein a gap is formed between the pair of guide slopes and the pair of tapered surfaces when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

19. The tire valve according to claim 15, wherein the guide part of the electronic device includes a tapered surface formed on surfaces of the pair of arcuate grooves facing each other and sloped toward the valve stem in a direction toward the retaining part, and the guide part of the support ring includes a tubular protrusion received between the pair of arcuate grooves, and a pair of guide slopes formed at two 180° spaced apart locations on the tubular protrusion and sloped toward each other in a direction toward the retaining part, the guide slopes being parallel to a third direction perpendicular to the first direction and the second direction, the pair of guide slopes and the tapered surfaces being adjacent to or abutting each other when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.

20. The tire valve according to claim 19, wherein a gap is formed between the pair of guide slopes and the pair of tapered surfaces when viewed from the third direction, when the support ring and the electronic device are set in the normal rotational position.
